

MUNTASIR SHOHRAB

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🌐 LinkedIn ◊ 🐙 GitHub ◊ 🎓 Scholar ◊ 🌐 Website

PROFESSIONAL SKILLS

ML & AI	Machine Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Speech Processing, Computer Vision, Multimodal Learning, Financial Machine Learning, Risk Modeling, Causal Inference, Time Series Modeling, Retrieval-Augmented Generation (RAG), Model Explainability (SHAP), Feature Engineering, Transformers
Models	LLaMA-3, Wav2Vec2, DeepLabv3+, BERT
Frameworks	PyTorch, CUDA, Hugging Face Transformers, PEFT/LoRA, scikit-learn, XG-Boost
Engineering	Modular Architecture, Agile Development, Software Development Lifecycle, Algorithms and Data Structures, Databases (SQLite, SQL), REST API Design, CLI Development, Deployment
Tools	Python, SQL, LaTeX, Linux, SLURM/HPC, Multi-GPU Training, Git, Playwright, OAuth2, Jinja2
Testing	Unit Testing (unittest/pytest), Test-Driven Development, Integration Testing, Code Review

EDUCATION

Ph.D., Business Data Science	September 2022 – Present
New Jersey Institute of Technology, Newark, New Jersey.	Current GPA: 3.96/4.0
M.Sc. in Applied Mathematics	2018 – 2019
University of Dhaka, Bangladesh.	GPA: 3.78/4.0
Thesis: “Predicting dam-break flow dynamics using Multi-Layer Perceptron (MLP).”	
B.Sc. in Applied Mathematics	2014 – 2018
University of Dhaka, Bangladesh.	GPA: 3.70/4.0

PROFESSIONAL WORK EXPERIENCE

Samsung R&D Institute	March 2018 – September 2018
<i>Intern</i>	<i>Dhaka, Bangladesh</i>
· Prototyped a semantic segmentation pipeline (DeepLabv3+, PyTorch) to classify UI-relevant image regions, improving auto-tagging accuracy by 5.2%. · Collaborated with senior engineers to design and prototype a CNN-based feature extraction module for an image viewer application. · Gained experience in applied computer vision, model inference optimization, and production-ready ML pipelines.	
MetLife	March 2017 – September 2017
<i>Intern</i>	<i>Dhaka, Bangladesh</i>
· Selected for a stipend-supported actuarial training program; completed structured prep in CT1 (Financial Mathematics—time value of money, annuities, bond pricing). · Built spreadsheet models for discounting, amortization, and bond cash flows; validated results against analytical solutions. · Gained practical exposure to actuarial workflows and controls.	

PROJECTS

- **Job Search Automation Agent** [GitHub]: Multi-agent CLI application with scraper agents (LinkedIn, RemoteOK), an LLM-based job scoring pipeline, browser automation (Playwright) for ATS form filling, Gmail OAuth integration for email classification, and a Notion database backend. End-to-end automation from job discovery to application submission with a human-in-the-loop approval gate.
- **ESG Visual Intelligence Pipeline (PPT-Analysis)** [GitHub]: Built a VLM-powered research pipeline to classify ESG/sustainability content across 500+ Fortune 500 corporate earnings presentations (14,000+ slides, 2010–2022). Automated slide classification using Google Gemini 2.5 Flash and GPT-4.1-mini; fine-tuned Qwen2-VL-2B with PEFT/LoRA for domain adaptation; performed event study analysis (Cumulative Abnormal Returns, Fama-French 3-factor model) to test whether visual ESG signals predict stock market reactions. End-to-end VLM pipeline from raw PDFs to financial econometrics at Fortune 500 scale.

- **NLP Mortality Prediction (NJACTS)** [GitHub]: Two-stage NLP pipeline: a configurable NoteTemplate engine serializes tabular EHR patient data (age, comorbidities, vitals) into natural language clinical notes, then fine-tunes GPT-2 for binary mortality prediction with class-weighted loss to handle label imbalance. Demonstrates table-to-text generation, transformer fine-tuning, and clinical NLP on real EHR data.
- **Reverse Visual Search (MS COCO)** [GitHub]: Reverse image search system over the MS COCO mini dataset: Faster R-CNN (ResNet-50 FPN) detects and crops all persons with >95% confidence, extracts 2048-dim pool-layer features, applies PCA, then retrieves the top-10 most similar persons via cosine similarity for any query crop. Demonstrates end-to-end computer vision pipeline: detection, feature extraction, dimensionality reduction, and similarity search.

ACADEMIC RESEARCH EXPERIENCE

Graduate Researcher

New Jersey Institute of Technology, Newark, New Jersey

September 2022 – Present

- **FinTabLLM**: Built a framework to serialize structured tabular M&A data to natural language and fine-tuned LLaMA-3 with parameter-efficient methods (PEFT) on multi-GPU infrastructure to predict merger withdrawals. Improved ROC AUC score by 2% compared to tree-based baselines.
- **Macroeconomic ML for Corporate Bond Risk**: Modeled interactions between bond characteristics and macroeconomic indicators using MLPs and linear regression to predict corporate bond excess returns. Out-of-sample R^2 score increased by 3% using a sparse curated feature set.
- **The Voice of Risk**: Developed verbal and acoustic ML features (nonanswers, disfluency) from CEO earnings calls using Wav2Vec2 and multinomial inverse regression to improve corporate misconduct prediction. Improved accuracy by 2.5%.
- **EPS Prediction with Multimodal Data**: Tokenized time-series and 10-Q reports; used RAG to extract forward-looking statements; predicted next-quarter EPS. Improved EPS prediction by 5% R^2 over time-series baseline.
- **Physics-Informed LLMs — Trigger Prediction at RHIC**: Converted collider simulation data into textual prompts; fine-tuned LLMs to classify high-energy physics trigger events. Demonstrated LLM applicability to physics event classification.

PUBLICATIONS & SELECTED PRESENTATIONS

- M. Shohrab, Z. Ye, Y. Liu, and D. Yu, “FinTabLLM: Unlocking the Power of LLMs for Merger Withdrawal Prediction,” *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, accepted, 2026.
- L. Chen, D. Zhou, M. Shohrab, X. Tao, L. Hao, T. Ma, and D. Yu, “Tensor-TimesFM: High-dimensional Sparse Time Series Forecasting with Tensor Decoders and Latent Dynamics Prediction,” *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, accepted, 2026.
- M. Shohrab, Z. Ye, Y. Liu, and D. Yu, “LLM-Based Merger Withdrawal Prediction and Model–Market Disagreement Around Climate Risk,” *Journal of Risk and Financial Management*, 19(7), 488, 2026.
- M. Z. I. Bangalee and M. Shohrab, “Determination of Pi Terms by the Method of Repeating Variables and Rayleigh Method: A Comparative Study,” *IAENG International Journal of Applied Mathematics*, March 2022.
- Z. Ye, M. Shohrab, Y. Liu, and D. Yu, “The Voice of Risk: Managerial Speech Cues and Corporate Misconduct,” *Financial Management Association (FMA) 2025 Annual Meeting*, Vancouver, accepted for presentation, 2025 and Revise & Resubmit (major revision), *Journal of Business Ethics*.
- M. Shohrab, Z. Ye, Y. Liu, and D. Yu, “Man vs. Machine: Unlocking the Power of LLM for Financial Risk Prediction Amid Climate Change,” *Southern Finance Association (SFA) 2025 Annual Meeting*, Orlando, accepted for presentation, 2025.
- Additional presentations: supply chain network centrality and abnormal returns (POMS 2025); heterogeneous M&A and innovation (EAM 2024); Buckingham Pi theorem (21st International Mathematics Conference).

AWARDS AND ACHIEVEMENTS

- Recipient, National Science and Technology (NST) Fellowship, Bangladesh (2021)
- Second Place, Mathcon (Math Olympiad), Bangladesh (2016)

REFERENCES

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